

NIKHIL NAIR

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Summary

Software engineer with hands-on experience building edge CV pipelines, robotics perception systems, and hardware-software integrations. Currently leading R&D at Super FabLab Kerala (MIT & KSUM), with a published patent, 4 research publications, and a B.Tech in Computer Science & Engineering. Open to embedded, SDE, or AI/ML engineering roles.

Experience

- Super FabLab Kerala – MIT & KSUM**  Nov 2025 – Present
Software Developer - Technology Fellow Kochi, India
- Cut manual setup overhead by **80%** by implementing a **computer vision pipeline** for automated bed calibration; achieved **0.2mm alignment precision** via real-time object localization on an embedded platform.
 - Reduced unit cost to **INR 2,000** (vs.~ INR 40,000 industrial alternatives) – a **95% cost reduction**, saving **INR 10L+** at scale, by engineering a custom **ESP32 Web-Plotter** with client-side G-code parsing (**60% MCU load reduction**).
 - Engineered **bLoom**, an **Edge AI** kinetic installation for **Huddle Global 2025**; optimized **Python/CV** logic to slash CPU overhead by **45%** and achieved **sub-50ms latency** via **Shell-based** headless automation.
- TATA Elxsi**  Oct 2023 – Dec 2023
AI and Robotics Intern (Transport Business Unit) Chennai, India
- Improved obstacle detection by fusing 3D LiDAR with 2D vision using **Python**, extending range in tough conditions.
 - Optimized KITTI dataset preprocessing by ~30% using **ROS2** and **OpenCV**, enhancing the perception pipeline.

Education

- Vellore Institute of Technology - Chennai** Sept 2021 - Sept 2025
B.Tech in Computer Science and Engineering (Cyber Physical Systems) CGPA: 8.21/10 or 3.3/4
- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Computer Networks, Software Development Lifecycle, Socket Programming, Artificial Intelligence, Core Java, Control Systems, UI/UX, LLMs.

Projects

- Urumi V2 — Camera-Assisted 4 Axis CNC Cutter**  MIT Collaboration | Jan 2026 – Present
- Leading camera module integration and sensor fusion on an MIT-collaboration CNC project; architecting a **lightweight modular UI**, communication and developing algorithms for **camera calibration** and automated material alignment for vision-guided cutting.
- NAS-Optimized Deep Learning Model for Concrete Strength Prediction**  Thesis | 2025
- Engineered a **Neural Architecture Search**-optimized model in **Python & Keras** for concrete strength prediction; improved accuracy by ~10% ($R^2 > 0.90$). Supported design of stronger, cost-effective, and sustainable mixes.

Technical Skills

Languages: Java, Python, C++, GoLang, SQL, PHP, HTML/CSS/JS, React.js, Node.js
Frameworks/Tech: Spring Boot, Docker, Flask, TensorFlow, REST API, OpenCV, ROS2, Jenkins
Tools/Platforms: Git, GitHub, Google Cloud Platform, Arduino, Overleaf, Figma, CI/CD, Jenkins, Gemini, Claude
Soft Skills: Communication, Team Collaboration, Problem Solving, Presentation, Analytical Thinking, Creativity

Patent & Publication

- A System to Recognize Number Plate and Allow Entry of a Vehicle**  2024
Patent Application Number: 202441051527 Published
- Comprehensive Review of Blockchain-Oriented Methods in Agricultural SCM**  2026
Harvesting Data: Blockchain, AI and Advanced Innovations in Agriculture Accepted
- Inverter-Integrated ECC w/ Torsion Points as Black Box Alternatives in Aircraft Security**  2025
Soft Computing Applications in Modern Power and Energy Systems Accepted

Certifications & Achievements

Certifications: Quantum Computing (Udemy, 2025) – covered **Python, Q#, Qiskit**; ML Specialization (Coursera, 2023); **Google Cloud** Career Practitioner & Foundation Programs (2023–24).
Achievements: **Top 25/548** in VITISH (SIH) Hackathon 2024; invited to pitch at VNEST incubator. Awarded **PMSS Scholarship** for academic excellence & leadership; **Open source contributor** - merged PRs in projects with **1K+ stars**.

Extracurricular

- Volunteer Teacher - U&I Trust** July 2024 – Apr 2025
- Led academic sessions for underprivileged students; supported a INR 2.13 crore education fundraiser through outreach.